MIL-B-52371A(EL)
14 January 1971
Superseding
MIL-B-52371(MO)
26 June 1964

MILITARY SPECIFICATION

BINOCULARS, ELECTRONIC

AN/PAS-5

1. SCOPE

1.1 This specification covers a headmounted, near infrared image forming binocular.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-781	- Steel, Strapping, Flat
PPP-B-585	- Box, Wood, Wirebound
PPP-B-601	- Box, Wood, Cleated-Plywood
PPP-B-621	- Box, Wood, Nailed and Lock Corner
PPP-B-636	- Box, Fiberboard
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, for Carton Sealing
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced

FSC 5855

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Military

MIL-B-18	- Batteries, Dry
MIL-P-116	- Preservation, Methods of
MIL-T-152	- Treatment, Moisture and Fungus Resistant, of Communications, Electronics and Associated Electrical Equipment
MIL-P-11268	- Part, Materials and Processes Used in Electronic Communications Equipment
MIL-E-55332	- Electron Tube, Image Converter, Type 8598

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U. S. Military Property
MIL-STD-147	- Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-810	- Environmental Test Methods

DRAWINGS

USAECOM

SC-DL-623799	- Binocular, Electronic

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction. - The binoculars shall be in accordance with SC-DL-623799 and as specified herein.

3.2 Preproduction. - The supplier shall furnish binoculars (see 6.2) within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods will produce binoculars that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish binoculars conforming to this specification.

3.3 Initial production. - When specified the supplier shall furnish to the Government binoculars (see 6.2) for inspection as specified in 4.4.

3.4 Materials and components. - Materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not specified shall be selected by the supplier and shall conform to MIL-P-11268.

3.4.1 Battery. - The battery shall be a type BA-1312/U per MIL-B-18. Unless otherwise specified, the battery shall be furnished by the contractor (see 6.2).

3.4.2 High voltage oscillator. - The high voltage oscillator shall be in accordance with SC-DL-623799.

3.4.3 Multiplier. - The multiplier shall be in accordance with SC-DL-623799. It shall be rigidly bonded to the image converter tube as a wrap-around component with the electrical connections permanently secured by a corona free weld or solder connection to the tube assembly. Electrical connections to the input and ground return of the multiplier shall be spring loaded non-corrosive material. The spring tension shall not suffer measurable fatigue after 500 tube replacement operations.

3.4.4 Electron tube, image converter. - The image converter tube shall be in accordance with MIL-E-85332.

3.5 Weight and dimensions.

3.5.1 Weight. - The binocular assembly, including head mount and mask shall weigh not more than 18 ounces. The complete assembly with carrying case and accessories shall weigh not more than 44 ounces.

3.5.2 Dimensions. - Dimensions shall be in accordance with SC-DL-623799, and as specified herein.

3.6 Operation and environmental.

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3.6.1 Modulation transfer function (MTF) objective lens. - The objective lens MTF shall be equal to or greater than the values listed below where LP/MM is line pairs per millimeter.

LP/MM AT PHOTOCATHODE	MTF VALUES IN %
20.0	55
40.0	40
60.0	25

3.6.2 Modulation transfer function (MTF) eyepiece lens. - The eyepiece lens MTF shall be equal to or greater than the values listed below:

LP/MM AT PHOTOCATHODE	MTF VALUES IN %
20.0	55
40.0	40
60.0	25

3.6.3 Objective focus. - The objective lens shall be focused and secured at infinity.

3.6.4 Focus adjustment-eyepiece. - The eyepiece focus adjustment shall insure an eyepiece focus adjustment range of plus 4, plus or minus 0.50, to minus 4, plus or minus 0.50, diopters by applying not more than 5.0 inch-pounds of torque.

3.6.5 Field of view. - The field of view shall be not less than 26 degrees.

3.6.6 Collimation. - The optical axis of each of the two monoculars of each operational binocular when focused at infinity and with an interpupillary distance between 55 and 72 mm shall be parallel within the following limits:

a. 12 minutes of arc of true field in a direction perpendicular to the horizontal plane of the line of sight.

b. 18 minutes of arc of true field in a direction perpendicular to the vertical plane of the line of sight.

3.6.7 Magnification and linear distortion. - A 6 mm diameter circular image, when projected on the center of the cathode, shall produce an image on the screen which measures 6.5 + 0.50 mm on any diameter.

3.6.8 High voltage breakdown. - The housing assembly, with the field-corrector lens bonded in place, shall withstand 15,000 volts between the inner and outer surfaces without voltage breakdown.

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3.6.9 Vibration. - The binocular shall not be damaged (see 6.3.1) when subjected to vibration specified in 4.6.9.

3.6.10 Immersion. - The binocular shall be watertight and shall not be damaged by immersion to a depth of 3 feet in fresh water for a period of 30 minutes.

3.6.11 Salt fog. - The binocular shall operate as specified herein and shall not be damaged after having been tested in accordance with Method 509 of MIL-STD-810.

3.6.12 Fungus resistance. - The binoculars (except for batteries, electrical components, such as switches, electrical contacts, light transmitting elements, and components hermetically sealed) shall be treated for fungus resistance in accordance with MIL-T-152, Type II.

3.6.13 Storage temperature.

3.6.13.1 High temperature storage. - The binoculars shall not be damaged by storage in any ambient temperature up to plus 68° C.

3.6.13.2 Low temperature storage. - The binoculars shall not be damaged by storage in any ambient temperature to minus 54° C.

3.6.14 Operational temperature.

3.6.14.1 High temperature operation. - The binoculars shall perform as specified herein at any ambient temperature to plus 52° C.

3.6.14.2 Low temperature operation. - The binoculars shall perform as specified herein at any temperature to minus 32° C.

3.6.15 Thermal shock. - The binoculars shall not be damaged by temperature shocks from plus 52° C to room temperature (see 6.3.2) and from minus 54° C to room temperature.

3.6.16 Humidity. - The binoculars shall not be damaged when exposed to a 90 to 98 percent relative humidity at temperatures from plus 29° C to plus 52° C.

3.6.17 Electromagnetic compatibility. - The electromagnetic compatibility of the binocular shall be in accordance with Table A-1-MIL-E-55301 of the Appendix to MIL-STD-461, Equipment Class Tactical-Non CE.

3.6.18 Mean time between failure. - The binoculars shall have a specified mean time between failure (MTBF) of 4000 hours.

3.7 Technical publications. - Such technical publications as are specified shall be furnished (see 6.2).

3.8 Identification markings. - Identification markings shall be in accordance with the applicable drawings and MIL-STD-130.

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3.9 Workmanship. - All parts, components, and assemblies of the binocular including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and electrical wiring and welded parts shall be clean and free from sand, dirt, pits, fins, sprues, scales, flux, and other harmful extraneous material. Joints and seams shall be tight, all edges shall be smoothly finished. The interior of the binocular assembly shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (see 4.3).

b. Initial production inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Inspection comparison (see 4.7).

e. Inspection of Preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. - Each preproduction binocular shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

MAJOR

101. Eyepiece diopter adjustment binds.

102. Does not operate (no characteristic green glow when turned on).

103. Broken optical or mechanical components.

104. External hardware on binocular loose.

MINOR

201. Technical publications not present as specified.

4.3.2 Tests. - Upon successful completion of the examination specified in 4.3.1 the preproduction models shall be subjected to the tests in Table I. Tests shall be conducted in the order listed. Failure of any test shall be cause for rejection.

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TABLE I - TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
High voltage breakdown	3.6.8	4.6.8
MTF objective lens	3.6.1	4.6.1
MTF eyepiece lens	3.6.2	4.6.2
Objective focus	3.6.3	4.6.3
Eyepiece focus	3.6.4	4.6.4
Collimation	3.6.6	4.6.6
Vibration	3.6.9	4.6.9
Magnification & linear distortion	3.6.7	4.6.7
Field of view	3.6.5	4.6.5
Low temperature storage	3.6.13.2	4.6.13.2
High temperature storage	3.6.13.1	4.6.13.1
Low temperature operation	3.6.14.2	4.6.14.2
High temperature operation	3.6.14.1	4.6.14.1
Thermal shock	3.6.15	4.6.15
Humidity	3.6.16	4.6.16
Salt fog	3.6.11	4.6.11
Immersion	3.6.10	4.6.10
Electromagnetic compatibility	3.6.17	4.6.17
MTF objective lens	3.6.1	4.6.1
MTF eyepiece lens	3.6.2	4.6.2
High voltage breakdown	3.6.8	4.6.8
Collimation	3.6.6	4.6.6
Mean time between failure	3.6.18	4.6.18

4.4 Initial production inspection. - When specified (see 3.3), initial production binoculars will be selected at random by the Government from the first binoculars produced from production tooling, shall be examined as specified in 4.3.1, and tested as specified in Table I to determine conformance to the requirements of this specification. The inspection will be performed by the contractor under supervision of a representative of the contracting officer. Acceptance of initial production binoculars shall not exclude the remaining binoculars from the quality conformance inspection and acceptance provisions specified in Section 4.

4.4.1 Inspection failure. - Failure of an initial production binocular to meet any requirement specified herein, during and as a result of the examination

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and tests specified in 4.4 shall be cause for rejection of the initial production binoculars and shall be cause for refusal by the Government to continue acceptance of production binoculars until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on binoculars previously produced and accepted under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all binoculars accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. - Each binocular shall be examined as specified in 4.3.1. One or more defects shall be cause for rejection of the unit.

4.5.2 Tests.

4.5.2.1 Group A . - Each binocular shall be tested in accordance with Table II. Failure of any test shall be cause for rejection of that unit.

TABLE II - GROUP "A" TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
MTF objective lens	3.6.1	4.6.1
MTF eyepiece lens	3.6.2	4.6.2
High voltage breakdown	3.6.8	4.6.8

4.5.2.2 Group B. - Group B tests shall be conducted on binoculars selected from units which have passed examination and tests in 4.5.1 and 4.5.2.1. Sampling shall be in accordance with MIL-STD-105, AQL 1.5 percent defective, inspection level S-3. Tests shall be conducted in accordance with Table III, and may be performed in any sequence except the vibration and the immersion tests shall be first and second in that order.

TABLE III - GROUP "B" TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Vibration	3.6.9	4.6.9
Immersion	3.6.10	4.6.10
Collimation	3.6.6	4.6.6
High voltage breakdown	3.6.8	4.6.8
Eyepiece focus	3.6.4	4.6.4

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TABLE III - GROUP "B" TEST SCHEDULE (Cont)

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Objective focus	3.6.3	4.6.3
MTF objective lens	3.6.1	4.6.1
MTF eyepiece lens	3.6.2	4.6.2

4.5.2.3 Group C. - This inspection shall consist of the tests specified in Table IV and shall be performed on units that have been subjected to and met Groups A and B inspection. Sample units shall be selected in accordance with 4.5.2.3.1. The tests shall be performed in the sequence listed.

4.5.2.3.1 Sampling for inspection. - Five (5) sample units shall be selected at random for inspections in Table IV, except for MTBF, from every 100 produced or every 30 days, whichever occurs first. A minimum of ten (10) units shall be tested for MTBF selected from units which have met the requirements of Group A. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot (see 4.5.2.3).

TABLE IV - GROUP "C" TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Humidity	3.6.16	4.6.16
High temperature storage	3.6.13.1	4.6.13.1
Low temperature storage	3.6.13.2	4.6.13.2
High temperature operation	3.6.14.1	4.6.14.1
Low temperature operation	3.6.14.2	4.6.14.2
Magnification & linear distortion	3.6.7	4.6.7
MTF objective lens	3.6.1	4.6.1
MTF eyepiece lens	3.6.2	4.6.2
Mean time between failure	3.6.18	4.6.18

4.5.2.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of the investigation and details of the proposed corrective action on (i) the processes and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Government Quality Assurance Representative. After corrective action has been taken,

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additional sample units shall be subjected to Group C inspection (all inspections, or the inspection which the unit failed, at the option of the Government) and Groups A and B inspection which may be reinstituted; however, final acceptance and shipment may be discontinued until the Group C reinspection results have shown that the corrective-action was effective.

4.5.2.3.3 Reinspection of group B and group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B and C inspections may be accepted on contract or order provided all damage is repaired and the units are resubjected to and pass group A inspection.

4.6 Test procedure. - Unless otherwise specified, all tests shall be performed at room temperature. Unless otherwise specified the binocular shall contain all elements necessary to functional operation. Test facilities, chambers and apparatus used in conducting tests specified herein shall conform to the requirements of MIL-STD-810.

4.6.1 Modulation transfer function - objective lens. - The objective lens shall be placed in a lens chuck or other suitable holding fixture of an MTF analyzer. Set the tungsten source for 2870° K plus or minus 50° K and focus the image of the slit onto the mask using a microscope objective lens specified for use with the MTF analyzer. Obtain a recording of the MTF including the three points specified in 3.6.1. Failure to equal or exceed the specified values of 3.6.1 shall constitute failure of this test.

4.6.2 Modulation transfer function - eyepiece. - Perform the eyepiece lens MTF test as specified in 4.6.1, except that the eyepiece lens assembly is used in place of the objective lens assembly. Failure to comply with the values specified in 3.6.2 shall constitute failure of this test.

4.6.3 Objective focus. - Using a suitable light source and collimator, a resolution pattern shall be projected on the photocathode resulting in 25 LP/MM when measured at the cathode. Failure of the monocular to resolve the pattern shall constitute failure of this test.

4.6.4 Focus adjustment - eyepiece. - The binocular shall be focused on a real or collimated image. Adjust the eyepiece to one extreme mechanical position. Position a diopter tester against the eyepiece lens and adjust until the image viewed through the binocular and diopter tester combination is in clear focus. Repeat the above procedure after adjusting the eyepiece lens for the other extreme mechanical position. Failure of the eyepiece to be adjustable from + 4 ± 0.5 diopters to -4 ± 0.5 diopters shall constitute failure of this test.

4.6.5 Field of view. - The field of view shall be examined with the binoculars by observing two vertical lines placed 55.4 inches apart on a vertical surface at a distance of 10 feet. This represents a field of view of 26 degrees. When both lines are simultaneously visible with the binocular held rigid in one position, the field of view is acceptable. A field of view less than 26 degrees shall constitute failure of this test.

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4.6.6 Collimation. - The binoculars shall be rigidly mounted in the collimator described in figure 1. The illuminated reticle used is that shown in figure 2. When the cross hairs of the telescope do not fall within the limits specified in 3.6.6 as indicated by the outline surrounding the cross hairs of the reticle, the locking ring shall be loosened and the three set screws adjusted until the cross hairs of the telescope and reticle coincide. Failure of the cross hairs to coincide within the limits specified in 3.6.6 shall constitute failure of this test.

4.6.7 Magnification and linear distortion. - By means of a collimator, a circle shall be projected onto the photocathode with the dimensions specified in 3.6.7. The diameter of the image of the circle appearing on the screen shall then be measured. Failure of the image to fall within the limits specified in 3.6.7 shall constitute failure of the test.

4.6.8 High voltage breakdown. - With the image-converter tube, eyepiece assembly and objective lens assembly removed from each monocular housing, a potential of 15,000 volts, dc shall be applied to a conductor placed in contact with the inside surface of the monocular housing and also in contact with the field-corrector lens inner seam, but insulated from both ground and high voltage contact springs. A potential probe of opposite polarity shall then be used to explore the complete outer surface of the housing with particular care taken to test the field-corrector lens outer seams. The probing end of the high-voltage probe shall consist of a ball having a minimum diameter of 1/4 inch and a maximum diameter of 5/16 inch. Arcing or other evidence of voltage breakdown shall constitute failure of this test.

4.6.9 Vibration. - Test the binocular in accordance with Method 201A of MIL-STD-202, except that the length of time for each mode shall be 15 minutes. The binoculars shall be securely mounted in place and held with metallic straps while being vibrated in each of the specified three mutually perpendicular directions. After each of the specified 15 minute periods, the binocular shall be examined for damage. Any evidence of damage shall constitute failure of this test.

4.6.10 Immersion. - With the binocular switch turned off, immerse the binoculars to a depth of 3 feet in fresh water for a period of 30 minutes. Remove the binoculars from the water and examine for:

a. Operation

b. Evidence of moisture in the power supply or binocular housing.

Failure to operate or evidence of moisture inside the housing shall constitute failure of this test.

4.6.11 Salt fog. - The binocular shall be tested in accordance with Method 509 of MIL-STD-810 using a 20 percent salt solution. Failure to pass the inspection of Method 509 shall constitute failure of this test.

4.6.12 Fungus resistance. - The binoculars, including case and accessories, shall be tested in accordance with Method 508 of MIL-STD-810. Operation is not required. Failure to pass the inspection of Method 508 shall constitute failure of this test.

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4.6.13 Storage temperature.

4.6.13.1 High temperature storage. - The binoculars shall be exposed to an ambient temperature of plus 68° C for a period of four (4) hours (see figure 3). At the conclusion of the 4 hour period inspect for damage. Any evidence of damage shall constitute failure of this test.

4.6.13.2 Low temperature storage. - The binoculars shall be exposed to an ambient temperature of minus 54° C for a period of 4 hours. At the conclusion of the 4 hour period inspect for damage. Any evidence of damage shall constitute failure of this test.

4.6.14 Operational temperature. -

4.6.14.1 High temperature operation. - The binoculars shall be exposed to an ambient temperature of plus 52° C for a period of 4 hours. At the conclusion of the 4 hour period the binoculars shall be turned on and inspected for damage and operation. Any evidence of damage or failure to operate shall constitute failure of this test.

4.6.14.2 Low temperature operation. - The binoculars shall be exposed to an ambient temperature of minus 32° C for a period of 4 hours. At the conclusion of the 4 hour period examine the binoculars for damage, mechanical operation and electrical operation (see figure 3). Any evidence of damage or failure to operate shall constitute failure of this test.

4.6.15 Thermal shock. - The binoculars shall be placed in a chamber in an ambient temperature of 52° C for a period of 4 hours. At the end of this period remove the binoculars from the chamber (in not more than 5 minutes) and allow to soak for 2 hours at room temperature. Examine for damage. Place the binoculars in a chamber with ambient temperature of minus 54° C and soak for 4 hours. At the end of this period remove the binoculars from the chamber (in not more than 5 minutes) and allow to soak at room temperature for 2 hours. Examine the binoculars for damage. Any evidence of damage shall constitute failure of this test.

4.6.16 Humidity. - Place the binocular in a chamber, with an internal temperature of plus 29° C and a relative humidity of 90 to 98 percent and soak for four hours. Raise the temperature of the chamber to 52° C in not more than 2 hours, maintain the relative humidity of 90 to 98 percent and soak for 4 hours. Lower the temperature of the chamber to 29° C in not more than 2 hours. This constitutes one complete cycle as shown in figure 4. Perform the above cycle six times. Turn the binocular on and observe the image tube operation for thelast 30 minutes of the 29°C portion of the 4th and 6th cycles. Failure to operate or evidence of damage during, or immediately following the above six cycles shall constitute failure of this test.

4.6.17 Electromagnetic compatibility. - The preproduction model, equipped for the reduction of electromagnetic interference in accordance with 3.6.17 shall be tested in accordance with MIL-STD-461, Tactical Support Class, Non-CE. The supplier shall furnish the contracting officer the test report and test plan required by MIL-STD-461 concurrent with the preproduction binocular. Upon approval of the report by the contracting officer and providing all other requirements of this specification are met, the preproduction model shall be used as a model for all production. Disapproval of the report shall constitute failure of this test.

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4.6.18 Mean time between failure. - Binocular testing shall be performed each 30 days on binoculars selected at random from the previous 30 day production period. The sample size shall be a minimum quantity of ten (10). For first article testing no replacements are permitted. For production testing, replacement of failed binoculars is required. Sample binoculars selected for testing shall be selected from lots which have passed the quality conformance inspection. The binoculars shall be operated at room temperature and the total number of operating hours for all units, failed and unfailed, original and replacement, shall be monitored and recorded. Once each 24 hours the binocular will be turned off for one hour, turned back on and the high voltage breakdown test performed (see 4.6.8). Observe the binocular for flickering, flashing, blanking, breakdown of any type or snapping or frying noises. Evidence of any of the above discrepancies or loss of operation shall constitute binocular failure. Failures shall be recorded and acceptability of the lot computed. The test shall be conducted in accordance with MIL-STD-781, Test Level A1, Test Plan VI. Failure to meet the above requirements shall constitute failure of the test. In the event of failure of the test 4.5.2.3.2 shall apply.

4.7 Inspection comparison. - The Procuring Contracting Officer may select binoculars at any time during the contract production period and subject these binoculars to the examination specified in 4.3.1 and the tests specified in Table I to determine that the quality of the selected binoculars is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Binoculars will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction or initial production binoculars.

4.7.1 Inspection failure. - Failure of any binocular when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of binoculars until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the failure.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. - For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. - Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. - Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105. Materials not as specified for level A or B.

106. Fiberboard boxes not reinforced with tape for level B.

107. Marking illegible, incomplete or incorrect.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. - The binocular shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - The binocular shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged as in method IC-1.

5.1.1.4.2 Binocular electronic AN/PAS-5. - The binocular shall be packaged method III as follows: Stow the accessories within the carrying case CY-6625/PAS-5. Stow the binoculars within the carrying case, close the case and secure fastenings. Cushion the case on all surfaces with cells or pads or both fabricated from fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned binocular within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature packaged as specified in 5.1.1.4.1, on top of the cushioned binocular directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. - The binocular shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2).

5.2.1 Level A.

5.2.1.1 Consolidation. - A quantity of binoculars packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

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5.2.1.2 Palletized load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I.

5.2.2 Level B.

5.2.2.1 Consolidation. - A quantity of binoculars packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1, shall be reinforced by the pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C .

5.2.3.1 Consolidation. - A quantity of binoculars packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard box shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. - These binoculars are intended for use as an aid in the night operation of various vehicles and equipment in areas equipped with infrared light sources and for the detection of infrared light sources.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. When preproduction or initial production tests are to be conducted and number of models of either or both which will be needed (see 3.2, 3.3, 4.3 and 4.4).

c. When the contractor shall furnish required batteries (see 3.4.1).

d. Technical publications to be furnished (see 3.7).

e. Level of preservation and packaging and the level of packing required (see 5.1 and 5.2).

f. Time interval required for submission of electromagnetic compatibility report (see 4.6.17).

6.3 Definitions.

6.3.1 Damage. - Damage is defined as breakage, loosening, failure, deterioration or corrosion of any finish, hardware, connection, or component; condensation of water within; or failure to operate in accordance with the provisions of this specification.

6.3.2 Room temperature. - Room temperature is defined as 23° C plus or minus 3°C.

Custodian:
Army-EL

Review Activity:
Army-ME,MU, AV, WC

Preparing Activity:
Army-EL

Project Number 5855-A067

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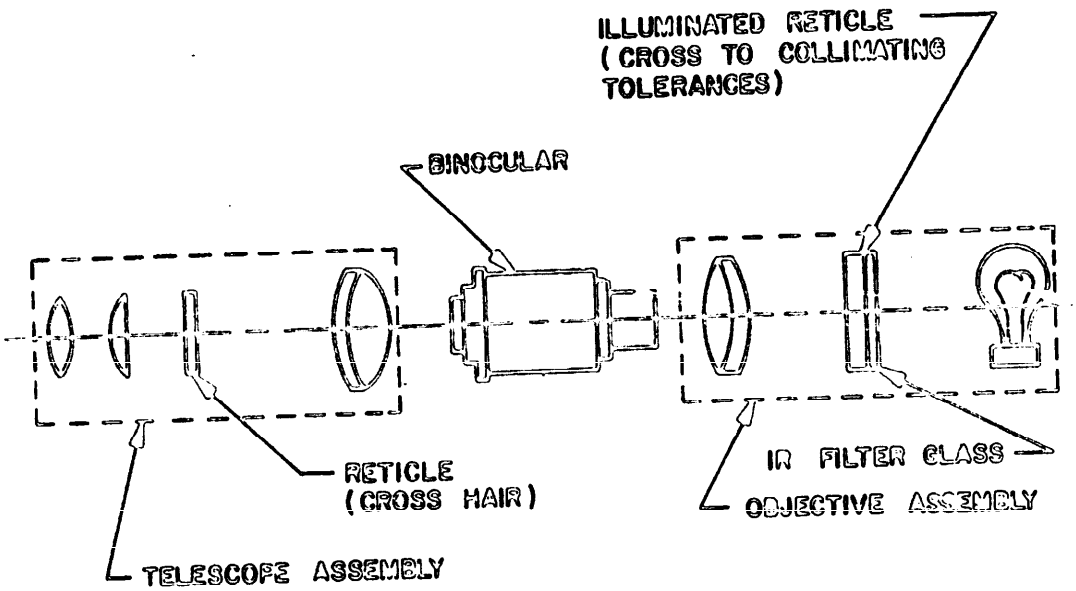


FIGURE 1. OPTICAL LAYOUT FOR INFRARED BINOCULAR COLLIMATOR (BINOCULAR, INFARARED)

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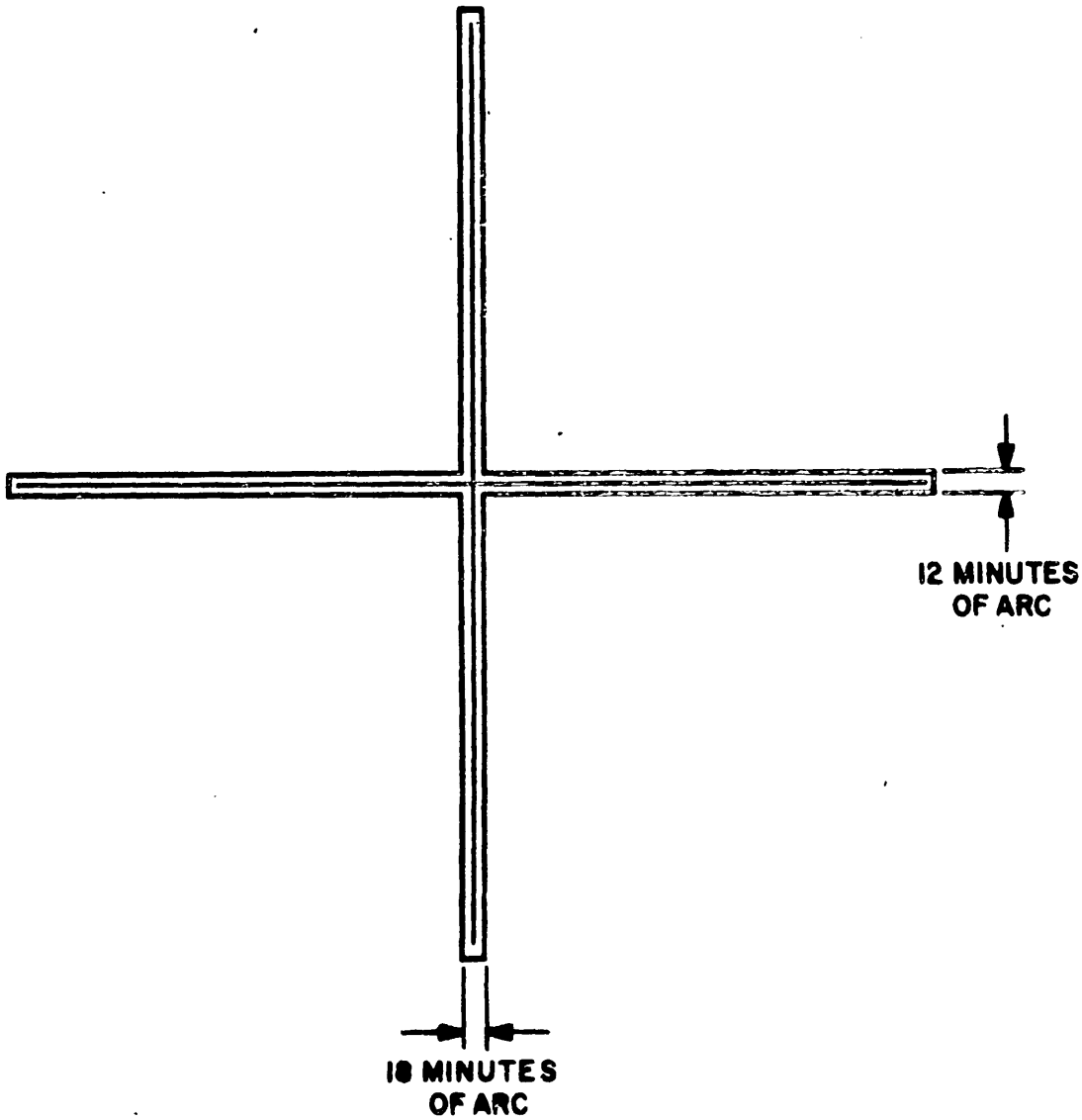


FIGURE 2. COLLIMATOR RETICLE (ILLUMINATED)

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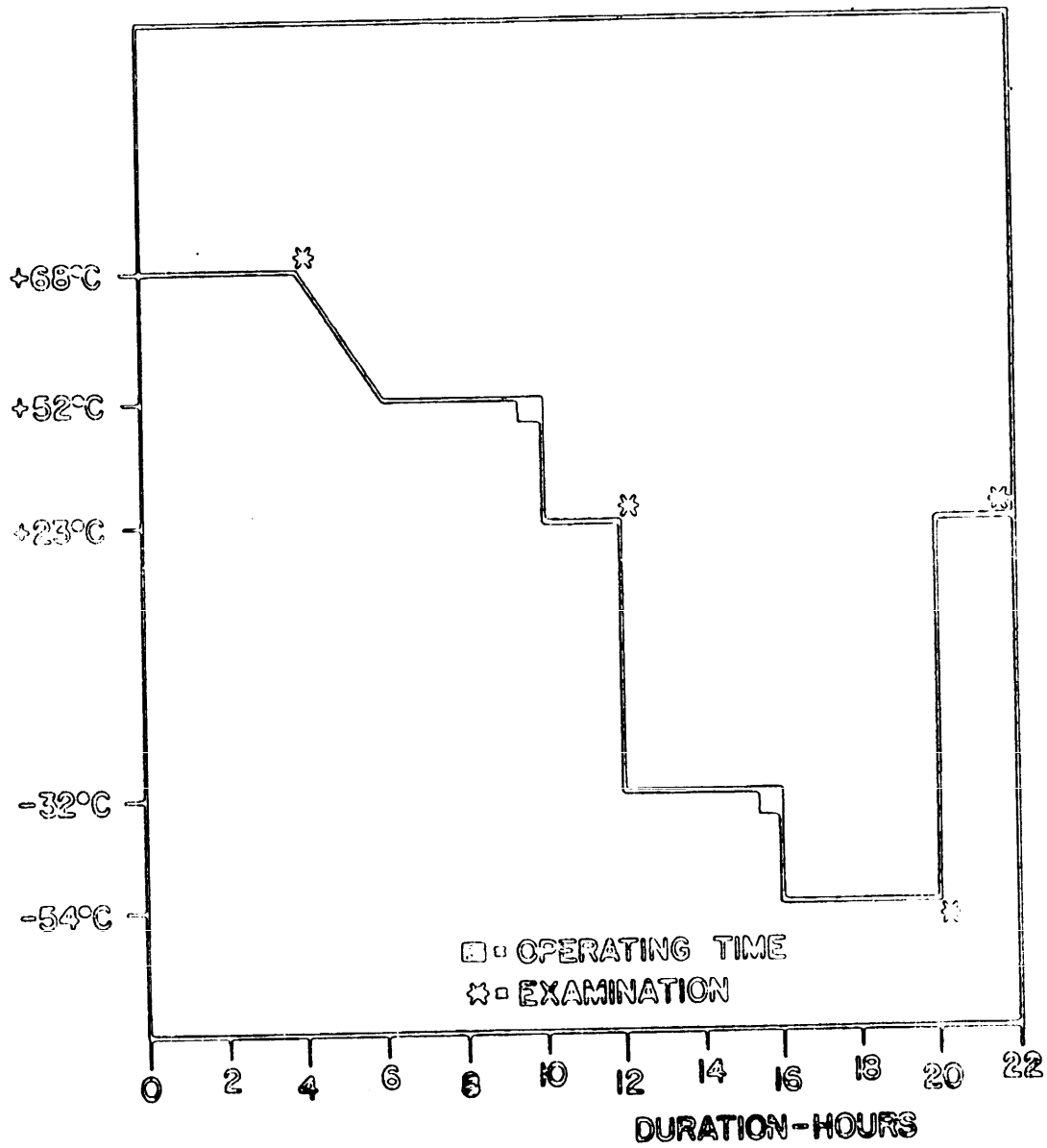


FIGURE 3. ENVIRONMENTAL CYCLE

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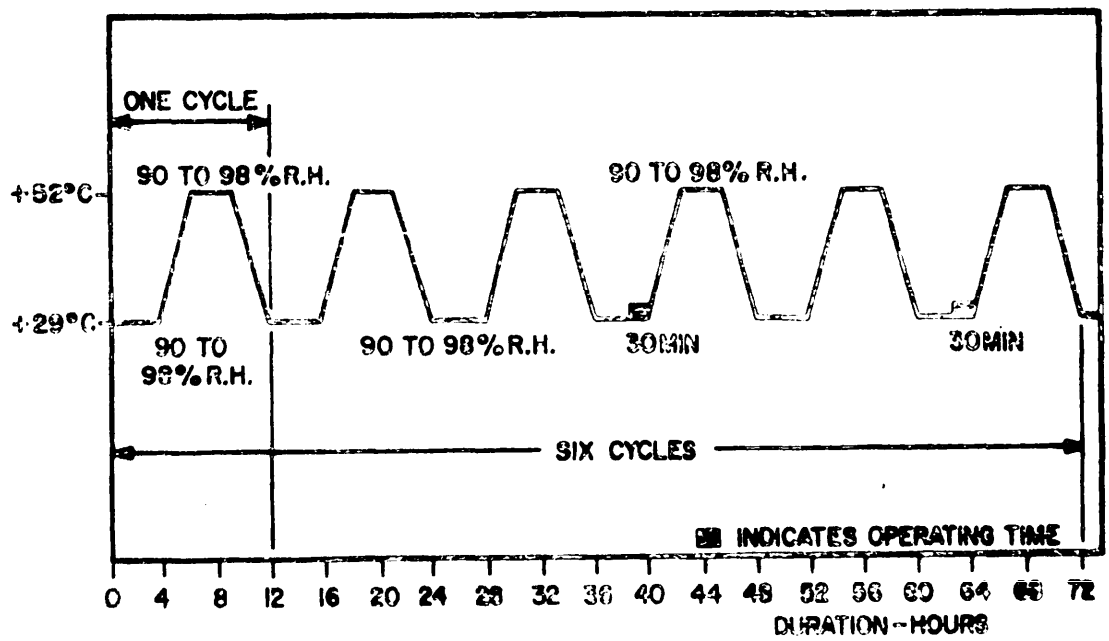


FIGURE 4. RELATIVE HUMIDITY TEST CYCLES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-B-52371-A(EL) BINOCULARS, ELECTRONIC, AN/PAS-5

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FN 1068-68

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